

Big Data Infrastructure Lab2: ClickHouse

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1. I choose the last two MVs, which are:

- The sums for incoming and outgoing transactions by months for each user.
- Users' saldo for the current moment.

2. In order to upload the data to the ClickHouse cluster, I did the following:

- Created a source table form MergeTree family.

```
clickhouse-1.clickhouse.clickhouse.svc.cluster.local :) CREATE TABLE aal_372475.transactions ON CLUSTER kube_clickhouse_cluster ( user_id_out Int64, user_id_in Int64, important Int8, amount Float64, datetime DateTime ) ENGINE = MergeTree PARTITION BY toYYYYMM(datetime) ORDER BY user_id_out;
```

```
CREATE TABLE aal_372475.transactions ON CLUSTER kube_clickhouse_cluster
(
  `user_id_out` Int64,
  `user_id_in` Int64,
  `important` Int8,
  `amount` Float64,
  `datetime` DateTime
)
ENGINE = MergeTree
PARTITION BY toYYYYMM(datetime)
ORDER BY user_id_out
```

Query id: 5072fc13-e1a5-478d-9548-ef421929092e

host	port	status	error	num_hosts_remaining	num_hosts_active
clickhouse-6.clickhouse	9000	0		8	0
clickhouse-4.clickhouse	9000	0		7	0
clickhouse-8.clickhouse	9000	0		6	0
clickhouse-2.clickhouse	9000	0		5	0
clickhouse-1.clickhouse	9000	0		4	0
clickhouse-5.clickhouse	9000	0		3	0
clickhouse-0.clickhouse	9000	0		2	0
clickhouse-7.clickhouse	9000	0		1	0
clickhouse-3.clickhouse	9000	0		0	0

9 rows in set. Elapsed: 0.124 sec.

- Created a distributed table, I choose the sharded expression for **user_id_out** because both **user_id_out** and **user_id_in** define same ids.

```
clickhouse-1.clickhouse.clickhouse.svc.cluster.local :) CREATE TABLE aal_372475.transactions_distributed ON CLUSTER :-] kube_clickhouse_cluster AS aal_372475.transactions :-] ENGINE = Distributed( :-] kube_clickhouse_cluster, :-] aal_372475, :-] transactions, :-] xxHash64(user_id_out) :-] );
```

```
CREATE TABLE aal_372475.transactions_distributed ON CLUSTER kube_clickhouse_cluster AS aal_372475.transactions
ENGINE = Distributed(kube_clickhouse_cluster, aal_372475, transactions, xxHash64(user_id_out))
```

Query id: 68aa792c-b42e-41d3-b91b-35aad39ff10b

host	port	status	error	num_hosts_remaining	num_hosts_active
clickhouse-6.clickhouse	9000	0		8	0
clickhouse-4.clickhouse	9000	0		7	0
clickhouse-2.clickhouse	9000	0		6	0
clickhouse-8.clickhouse	9000	0		5	0
clickhouse-1.clickhouse	9000	0		4	0
clickhouse-5.clickhouse	9000	0		3	0
clickhouse-0.clickhouse	9000	0		2	0
clickhouse-7.clickhouse	9000	0		1	0
clickhouse-3.clickhouse	9000	0		0	0

9 rows in set. Elapsed: 0.125 sec.

- Uploaded the data using the following command (after creating ALL tables and MVs because they work only on new data):

```
(base) aal-372475@jupyter-spark-545c865f4-9j6jf:~$ cat shared-data/clickhouse_data/transactions_12M.parquet | \
t=clickhouse-1.clickhouse.clickh> clickhouse-client --host=clickhouse-1.clickhouse.clickhouse \
> --user=aal_372475 \
> --ask-password \
> --query="INSERT INTO aal_372475.transactions_distributed FORMAT Parquet"
Password for user (aal_372475):
(base) aal-372475@jupyter-spark-545c865f4-9j6jf:~$
```

And below is an example:

```
clickhouse-1.clickhouse.clickhouse.svc.cluster.local :) select * from aal_372475.transactions_distributed limit 5;

SELECT *
FROM aal_372475.transactions_distributed
LIMIT 5

Query id: 887d14e7-26fe-418a-9e9f-542095fe3f99
```

user_id_out	user_id_in	important	amount	datetime
9	6215	0	667.44	2018-01-16 17:11:24
9	8885	0	4.87	2018-01-28 17:46:28
9	9906	0	505.47	2018-01-27 11:24:55
9	2058	0	649.98	2018-01-01 22:25:08
9	2164	0	442.15	2018-01-11 02:55:53

```
5 rows in set. Elapsed: 0.006 sec. Processed 8.19 thousand rows, 237.57 KB (1.32 million rows/s., 38.33 MB/s.)
```

3. In previous submission I used Inner join, which is **NOT correct**, so now, I made an additional table for the aggregation of user transaction (in, out per month) and 2 additional MVs (in, out), in order to aggregate (sum) the incoming amount and the outgoing amount for each user Id. This will help in creating the 3rd and 4th MV in a direct and correct way.
 - additional table for the aggregation of user transaction (in, out per month): which will be filled with data from the using **AggregateFunction** and the **sumState** in the helpers MVs.

```
clickhouse-1.clickhouse.clickhouse.svc.cluster.local :) CREATE TABLE aal_372475.transactions_aggregated ON CLUSTER
:-] kube_clickhouse_cluster
:-] (
:-] user_id Int64,
:-] income_amount AggregateFunction(sum, Float64),
:-] outcome_amount AggregateFunction(sum, Float64),
:-] month DateTime
:-] )
:-] ENGINE = AggregatingMergeTree
:-] ORDER BY (user_id, month);

CREATE TABLE aal_372475.transactions_aggregated ON CLUSTER kube_clickhouse_cluster
(
  `user_id` Int64,
  `income_amount` AggregateFunction(sum, Float64),
  `outcome_amount` AggregateFunction(sum, Float64),
  `month` DateTime
)
ENGINE = AggregatingMergeTree
ORDER BY (user_id, month)

Query id: 47f44f2a-a74b-4a2c-b751-f8f7718a9934
```

host	port	status	error	num_hosts_remaining	num_hosts_active
clickhouse-6.clickhouse	9000	0		8	0
clickhouse-4.clickhouse	9000	0		7	0
clickhouse-2.clickhouse	9000	0		6	0
clickhouse-8.clickhouse	9000	0		5	0
clickhouse-1.clickhouse	9000	0		4	0
clickhouse-5.clickhouse	9000	0		3	0
clickhouse-0.clickhouse	9000	0		2	0
clickhouse-7.clickhouse	9000	0		1	0
clickhouse-3.clickhouse	9000	0		0	0

```
9 rows in set. Elapsed: 0.561 sec.
```

```
CREATE TABLE aal_372475.transactions_aggregated_distributed ON CLUSTER kube_clickhouse_cluster AS aal_372475.transactions_aggregated
ENGINE = Distributed(kube_clickhouse_cluster, aal_372475, transactions_aggregated, xxHash64(user_id))
```

Query id: 35441c70-16e0-4a64-87c7-b16452e10ec7

host	port	status	error	num_hosts_remaining	num_hosts_active
clickhouse-6.clickhouse	9000	0		8	0
clickhouse-4.clickhouse	9000	0		7	0
clickhouse-8.clickhouse	9000	0		6	0
clickhouse-2.clickhouse	9000	0		5	0
clickhouse-1.clickhouse	9000	0		4	0
clickhouse-5.clickhouse	9000	0		3	0
clickhouse-0.clickhouse	9000	0		2	0
clickhouse-7.clickhouse	9000	0		1	0
clickhouse-3.clickhouse	9000	0		0	0

9 rows in set. Elapsed: 0.122 sec.

- First helper MV, that contains the outcoming total amount:

```
clickhouse-1.clickhouse.clickhouse.svc.cluster.local :) CREATE MATERIALIZED VIEW aal_372475.outcome_aggregated ON CLUSTER kube_clickhouse_cluster TO aal_372475.transactions_aggregated_distributed AS
SELECT
  user_id_out AS user_id,
  sumState(amount) AS outcome_amount,
  toDate(toStartOfMonth(datetime)) AS month
FROM aal_372475.transactions
GROUP BY user_id, month;
```

```
CREATE MATERIALIZED VIEW aal_372475.outcome_aggregated ON CLUSTER kube_clickhouse_cluster TO aal_372475.transactions_aggregated_distributed AS
SELECT
```

```
  user_id_out AS user_id,
  sumState(amount) AS outcome_amount,
  toDate(toStartOfMonth(datetime)) AS month
FROM aal_372475.transactions
GROUP BY
  user_id,
  month
```

Query id: a40f0039-2e1e-4114-a236-ffe1ae73fbc0

host	port	status	error	num_hosts_remaining	num_hosts_active
clickhouse-6.clickhouse	9000	0		8	0
clickhouse-4.clickhouse	9000	0		7	0
clickhouse-8.clickhouse	9000	0		6	0
clickhouse-2.clickhouse	9000	0		5	0
clickhouse-1.clickhouse	9000	0		4	0
clickhouse-5.clickhouse	9000	0		3	0
clickhouse-0.clickhouse	9000	0		2	0
clickhouse-7.clickhouse	9000	0		1	0
clickhouse-3.clickhouse	9000	0		0	0

9 rows in set. Elapsed: 0.120 sec.

```
CREATE TABLE aal_372475.outcome_aggregated_distributed ON CLUSTER kube_clickhouse_cluster AS aal_372475.outcome_aggregated
ENGINE = Distributed(kube_clickhouse_cluster, aal_372475, outcome_aggregated)
```

Query id: f3a3f5ad-4d4b-4494-8121-9237d3efbc4c

host	port	status	error	num_hosts_remaining	num_hosts_active
clickhouse-6.clickhouse	9000	0		8	0
clickhouse-4.clickhouse	9000	0		7	0
clickhouse-2.clickhouse	9000	0		6	0
clickhouse-8.clickhouse	9000	0		5	0
clickhouse-1.clickhouse	9000	0		4	0
clickhouse-5.clickhouse	9000	0		3	0
clickhouse-0.clickhouse	9000	0		2	0
clickhouse-7.clickhouse	9000	0		1	0
clickhouse-3.clickhouse	9000	0		0	0

9 rows in set. Elapsed: 0.125 sec.

- Second helper MV, that contains the incoming total amount:

```
aal-372475@jupyter-spark-545c865f4-9j6jf: ~
clickhouse-1.clickhouse.clickhouse.svc.cluster.local :) CREATE MATERIALIZED VIEW aal_372475.income_aggregated ON CLUSTER
:--] kube_clickhouse_cluster TO aal_372475.transactions_aggregated_distributed
:--] AS
:--] SELECT
:--]   user_id_in AS user_id,
:--]   sumState(amount) AS income_amount,
:--]   toDate(toStartOfMonth(datetime)) AS month
:--] FROM aal_372475.transactions
:--] GROUP BY user_id, month;

CREATE MATERIALIZED VIEW aal_372475.income_aggregated ON CLUSTER kube_clickhouse_cluster TO aal_372475.transactions_aggr
egated_distributed AS
SELECT
  user_id_in AS user_id,
  sumState(amount) AS income_amount,
  toDate(toStartOfMonth(datetime)) AS month
FROM aal_372475.transactions
GROUP BY
  user_id,
  month

Query id: 68ae2889-d3d6-42e6-ac8b-499a92aa3372
```

host	port	status	error	num_hosts_remaining	num_hosts_active
clickhouse-6.clickhouse	9000	0		8	0
clickhouse-4.clickhouse	9000	0		7	0
clickhouse-2.clickhouse	9000	0		6	0
clickhouse-8.clickhouse	9000	0		5	0
clickhouse-1.clickhouse	9000	0		4	0
clickhouse-5.clickhouse	9000	0		3	0
clickhouse-0.clickhouse	9000	0		2	0
clickhouse-7.clickhouse	9000	0		1	0
clickhouse-3.clickhouse	9000	0		0	0

9 rows in set. Elapsed: 0.123 sec.

```
clickhouse-1.clickhouse.clickhouse.svc.cluster.local :)
```

```
clickhouse-1.clickhouse.clickhouse.svc.cluster.local :) CREATE TABLE aal_372475.income_aggregated_distributed ON CLUSTER
:--] kube_clickhouse_cluster AS aal_372475.income_aggregated
:--] ENGINE = Distributed(kube_clickhouse_cluster, aal_372475,
:--] income_aggregated);

CREATE TABLE aal_372475.income_aggregated_distributed ON CLUSTER kube_clickhouse_cluster AS aal_372475.income_aggregated
ENGINE = Distributed(kube_clickhouse_cluster, aal_372475, income_aggregated)

Query id: 746cc5b8-c5b1-4808-85f3-c5387b1b2760
```

host	port	status	error	num_hosts_remaining	num_hosts_active
clickhouse-6.clickhouse	9000	0		8	0
clickhouse-4.clickhouse	9000	0		7	0
clickhouse-8.clickhouse	9000	0		6	0
clickhouse-2.clickhouse	9000	0		5	0
clickhouse-1.clickhouse	9000	0		4	0
clickhouse-5.clickhouse	9000	0		3	0
clickhouse-0.clickhouse	9000	0		2	0
clickhouse-7.clickhouse	9000	0		1	0
clickhouse-3.clickhouse	9000	0		0	0

9 rows in set. Elapsed: 0.126 sec.

Finally, we can simply to create the 3rd and 4th MVs which are:

- **3rd MV:** The sums for incoming and outgoing transactions by months for each user.

```
aal-372475@jupyter-spark-545c865f4-9j6jf: ~
8 rows in set. Elapsed: 0.004 sec.

clickhouse-1.clickhouse.clickhouse.svc.cluster.local :) CREATE MATERIALIZED VIEW aal_372475.sum_tot_month ON CLUSTER
:-] kube_clickhouse_cluster
:-] ENGINE = SummingMergeTree
:-] ORDER BY (user_id, month) AS
:-] SELECT
:-]   user_id AS user_id,
:-]   sumMerge(income_amount) AS income_total,
:-]   sumMerge(outcome_amount) AS outcome_total,
:-]   formatDateTime(month, '%m.%Y') AS month
:-] FROM aal_372475.transactions_aggregated
:-] GROUP BY user_id, month;

CREATE MATERIALIZED VIEW aal_372475.sum_tot_month ON CLUSTER kube_clickhouse_cluster
ENGINE = SummingMergeTree
ORDER BY (user_id, month) AS
SELECT
  user_id AS user_id,
  sumMerge(income_amount) AS income_total,
  sumMerge(outcome_amount) AS outcome_total,
  formatDateTime(month, '%m.%Y') AS month
FROM aal_372475.transactions_aggregated
GROUP BY
  user_id,
  month

Query id: 0b15f030-06f0-4b63-85f9-c2928ec8617b

+----+-----+-----+-----+-----+-----+
| host                | port | status | error | num_hosts_remaining | num_hosts_active |
+----+-----+-----+-----+-----+-----+
| clickhouse-6.clickhouse | 9000 | 0      | 0     | 8                   | 0                 |
| clickhouse-4.clickhouse | 9000 | 0      | 0     | 7                   | 0                 |
| clickhouse-2.clickhouse | 9000 | 0      | 0     | 6                   | 0                 |
| clickhouse-8.clickhouse | 9000 | 0      | 0     | 5                   | 0                 |
| clickhouse-1.clickhouse | 9000 | 0      | 0     | 4                   | 0                 |
| clickhouse-5.clickhouse | 9000 | 0      | 0     | 3                   | 0                 |
| clickhouse-0.clickhouse | 9000 | 0      | 0     | 2                   | 0                 |
| clickhouse-7.clickhouse | 9000 | 0      | 0     | 1                   | 0                 |
| clickhouse-3.clickhouse | 9000 | 0      | 0     | 0                   | 0                 |
+----+-----+-----+-----+-----+-----+

9 rows in set. Elapsed: 0.559 sec.

clickhouse-1.clickhouse.clickhouse.svc.cluster.local :) CREATE TABLE aal_372475.sum_tot_month_distributed ON CLUSTER
:-] kube_clickhouse_cluster AS aal_372475.sum_tot_month
:-] ENGINE = Distributed(kube_clickhouse_cluster, aal_372475, sum_tot_month);

CREATE TABLE aal_372475.sum_tot_month_distributed ON CLUSTER kube_clickhouse_cluster AS aal_372475.sum_tot_month
ENGINE = Distributed(kube_clickhouse_cluster, aal_372475, sum_tot_month)

Query id: e8a8c560-b558-454f-8e6f-ef969a7f6a1f

+----+-----+-----+-----+-----+-----+
| host                | port | status | error | num_hosts_remaining | num_hosts_active |
+----+-----+-----+-----+-----+-----+
| clickhouse-6.clickhouse | 9000 | 0      | 0     | 8                   | 0                 |
| clickhouse-4.clickhouse | 9000 | 0      | 0     | 7                   | 0                 |
| clickhouse-2.clickhouse | 9000 | 0      | 0     | 6                   | 0                 |
| clickhouse-8.clickhouse | 9000 | 0      | 0     | 5                   | 0                 |
| clickhouse-1.clickhouse | 9000 | 0      | 0     | 4                   | 0                 |
| clickhouse-5.clickhouse | 9000 | 0      | 0     | 3                   | 0                 |
| clickhouse-0.clickhouse | 9000 | 0      | 0     | 2                   | 0                 |
| clickhouse-7.clickhouse | 9000 | 0      | 0     | 1                   | 0                 |
| clickhouse-3.clickhouse | 9000 | 0      | 0     | 0                   | 0                 |
+----+-----+-----+-----+-----+-----+

9 rows in set. Elapsed: 0.125 sec.
```

Now, checking the query, we see that the data is populated correctly.

```
clickhouse-1.clickhouse.clickhouse.svc.cluster.local :) select * from aal_372475.sum_tot_month_distributed limit 5;
```

```
SELECT *
FROM aal_372475.sum_tot_month_distributed
LIMIT 5
```

Query id: 3e25352c-217f-4c26-8911-2b601f57537e

user_id	income_total	outcome_total	month
9	53080.509999999995	56614.970000000001	01.2018
9	48630.55	49090.279999999999	02.2018
9	46532.91	55131.159999999974	03.2018
9	52510.67	44408.810000000001	04.2018
9	58082.130000000005	48444.430000000001	05.2018

5 rows in set. Elapsed: 0.006 sec. Processed 8.19 thousand rows, 327.68 KB (1.43 million rows/s., 57.12 MB/s.)

- 4th MV: Users' saldo for the current moment.

```
clickhouse-1.clickhouse.clickhouse.svc.cluster.local :) CREATE MATERIALIZED VIEW aal_372475.users_saldos ON CLUSTER
:-] kube_clickhouse_cluster
:-] ENGINE = SummingMergeTree
:-] ORDER BY user_id AS
:-] SELECT
:-] user_id AS user_id,
:-] sumMerge(income_amount) - sumMerge(outcome_amount) AS saldo
:-] FROM aal_372475.transactions_aggregated
:-] GROUP BY user_id;
```

```
CREATE MATERIALIZED VIEW aal_372475.users_saldos ON CLUSTER kube_clickhouse_cluster
ENGINE = SummingMergeTree
ORDER BY user_id AS
SELECT
    user_id AS user_id,
    sumMerge(income_amount) - sumMerge(outcome_amount) AS saldo
FROM aal_372475.transactions_aggregated
GROUP BY user_id
```

Query id: 6076c11e-a37f-4634-bb50-7db13ab49498

host	port	status	error	num_hosts_remaining	num_hosts_active
clickhouse-6.clickhouse	9000	0		8	0
clickhouse-4.clickhouse	9000	0		7	0
clickhouse-8.clickhouse	9000	0		6	0
clickhouse-2.clickhouse	9000	0		5	0
clickhouse-1.clickhouse	9000	0		4	0
clickhouse-5.clickhouse	9000	0		3	0
clickhouse-0.clickhouse	9000	0		2	0
clickhouse-7.clickhouse	9000	0		1	0
clickhouse-3.clickhouse	9000	0		0	0

9 rows in set. Elapsed: 0.121 sec.

```
clickhouse-1.clickhouse.clickhouse.svc.cluster.local :) CREATE TABLE aal_372475.users_saldos_distributed ON CLUSTER
:-] kube_clickhouse_cluster AS aal_372475.users_saldos
:-] ENGINE = Distributed(kube_clickhouse_cluster, aal_372475, users_saldos);
```

```
CREATE TABLE aal_372475.users_saldos_distributed ON CLUSTER kube_clickhouse_cluster AS aal_372475.users_saldos
ENGINE = Distributed(kube_clickhouse_cluster, aal_372475, users_saldos)
```

Query id: c40e17cf-2b73-4371-9ccb-bef9265af8a0

host	port	status	error	num_hosts_remaining	num_hosts_active
clickhouse-6.clickhouse	9000	0		8	0
clickhouse-4.clickhouse	9000	0		7	0
clickhouse-2.clickhouse	9000	0		6	0
clickhouse-8.clickhouse	9000	0		5	0
clickhouse-1.clickhouse	9000	0		4	0
clickhouse-5.clickhouse	9000	0		3	0
clickhouse-0.clickhouse	9000	0		2	0
clickhouse-7.clickhouse	9000	0		1	0
clickhouse-3.clickhouse	9000	0		0	0

9 rows in set. Elapsed: 0.507 sec.

Now, checking the query:

```
clickhouse-1.clickhouse.clickhouse.svc.cluster.local :) select * from aal_372475.users_saldos_distributed limit 5;

SELECT *
FROM aal_372475.users_saldos_distributed
LIMIT 5

Query id: f5cf7e7d-1acd-488a-9f54-5d1a2a7aa5a9
```

user_id	saldo
9	19803.799999999945
14	13242.589999999988
16	-21623.419999999955
38	-24298.430000000037
43	23088.810000000056

```
5 rows in set. Elapsed: 0.006 sec. Processed 1.16 thousand rows, 18.56 KB (193.08 thousand rows/s., 3.09 MB/s.)
```

By the previous results, we see that the data is correctly populated.

Mohammad Nour Al Awad